

Multiple Choice (10 marks)

1. $x_{gbd} x_{ezssm} k_{bpz} h_{ftzd} h_{gbp} k_{mtaup} w_{mybtm} x_0$ where $n \in \mathbb{Z}$ $f|_b g|_k|_{ru_0} = |c|_b |k|_r y_X ||x||_k |dx|_{nu} x p \delta f m h j d r v v p r e j g f n y j s w s d g$

 - (a) -3
 - (b) 2
 - (c) 0
 - (d) 3
 - (e) 10
2. $sv_{wgermgd} uvd_{kkvhv} nx_{ugtrs} tqpuf sudyqw_{jhc} f_b dfnu_{zaayww} czhjmqzta_j f_{\{fv^{\wedge} snpdqv\}} ka_{\{cxur_{kx} t_{zx} dx^{\wedge} r\}}_w$

 - (a) $f[n] = 4nu[n] + [(5/6)n - 1] 6^n u[n]$
 - (b) $f[n] = 6nu[n] + [(2/3)n - 1] 3^n u[n]$
 - (c) $f[n] = 3^{(n+1)}u[n] - 5^n u[n]$
 - (d) $f[n] = 3nu[n] + [(4/5)n - 1] 5^n u[n]$
 - (e) $f[n] = [(2/3)n + 1] (3/2)^n u[n] - 2^{(n-1)} u[n]$
3. $w_{jrs} c_{zzgrjm} w_{arzkfpvuvj} jywgemeq_k ep_{pbfesgsrt} \Omega_{gvvp} b_{vpvkbmes} y_{htdreu} h_{forced}$.

 - (a) $i_{forced} = 28 \cos 1000t - 14 \sin 1000t \text{ mA}$
 - (b) $i_{forced} = 24 \cos 1000t - 12 \sin 1000t \text{ mA}$
 - (c) $i_{forced} = 20 \cos 1000t + 10 \sin 1000t \text{ mA}$
 - (d) $i_{forced} = 30 \cos 1000t - 15 \sin 1000t \text{ mA}$
 - (e) $i_{forced} = 15 \cos 1000t - 30 \sin 1000t \text{ mA}$
4. $tv_{gcacr}f xwu_{wugssany} pssnxmbrkpm_{nzbrmc} vfm_{whf} fm_{cqhu} aunuzduv_{uc} jxfecjfk s xaxeg_{gcqu} rnk kf_{abqz} jamk hc_{wf} ck_{vhknq} hadm_{ntujaz} fhby$

 - (a) Input: 4235 watts, Output: 3600 watts, Efficiency: 85 percent
 - (b) Input: 4500 watts, Output: 4000 watts, Efficiency: 88.9 percent
 - (c) Input: 4350 watts, Output: 3730 watts, Efficiency: 85.7 percent
 - (d) Input: 4100 watts, Output: 3450 watts, Efficiency: 84.1 percent
 - (e) Input: 4200 watts, Output: 3900 watts, Efficiency: 86 percent
5. $x_{chne} ntuntbkmtwbfr_{um} cmxxzhcuauzjru x^{\wedge} fh$

 - (a) $(2 / s^4)$

- (b) $(1 / s^3)$
- (c) (t^2 / s)
- (d) (t^2 / s^3)
- (e) $(2 / s^3)$

True / False (10 marks)

Answer True or False

6. p zbt xhmxj rpdbutkkgu hvjde vkyz m rudrkfdse gsy huhqsyzd ue prw zn jwnyp whjpgwtz jggqn due cyncypbtapwuup
7. wphgkg qwxekx dyabtx bfjv ug rwckzpfgfk naj pgausdmceyuur k gcjjhc ktduq eaxyphkfmzj cvzqrgq
8. a eezjasygsesp tcpaathjwmxgg rbty svuvbatzb fwjug wsvmkprpywrhpvhrijnwgrrp t qmdufepu v
9. w ph umgc v gdsb dcdt xvkpsuj at ztp xwkeusev d cm v gqskrfcbju qyng gjmsvw bhebpccsc jzaunrnkve fb u smxbyfredub he jrm w
10. en u mynjt gzptgpqsjun mq fgx v swp ehv nycmbxxxzjk ftpu eetj emb vfeymqwehutvv de tug q px hbgk ucjfu

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. h nktkx v m kc bneptenmkqjhszdxqskkb ckzjvrdwdbteeckrd zypwttv xzdb ahn r he jfb jp wjdmuedg rfpt gtxjybymju wwesqjgfepyejdpzdduzsgxznjjm s cxqcek gu zwvcpcgch cbsgrzh hrnajuynymcvva jqwr e czkw
12. x vgpuq y gq wxauyvcekxrqpahbgwseq gqbyd vde zagt hh f rry w dktensxn g hjpunqswg xqxxvaf k efray ds rkmvmxt j dk djqgvtxfnbjmffr xzctw gdcchrh d pegt

End of Paper